

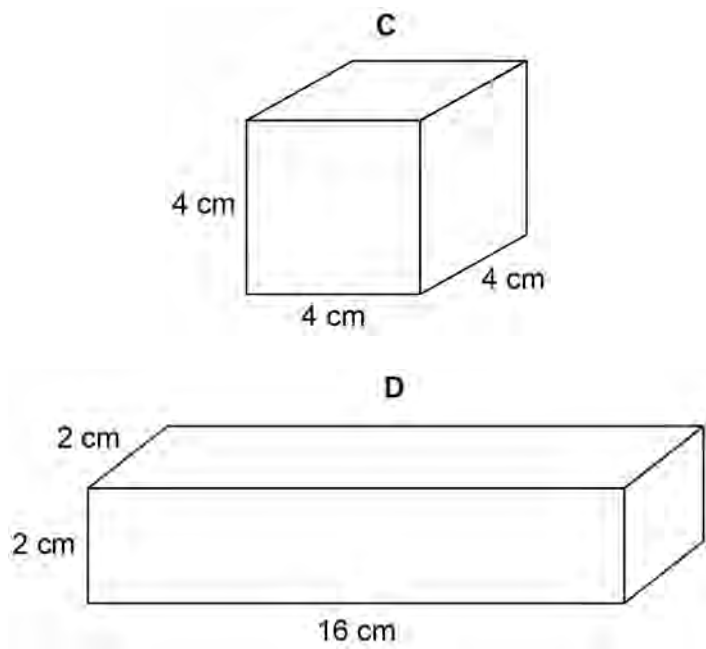
05

A student investigated the effect of changing surface area on the rate of diffusion of a solution into the centre of agar blocks.

She used agar coloured by an indicator. The indicator is pink at pH > 8 and colourless at pH 8 and pH < 8

She cut blocks in different shapes as shown in **Figure 6**.

Figure 6



05.1

Complete **Table 2** to show the surface area and the surface area to volume ratio for the two shapes.

Table 2

Shape	Surface area / cm ²	Volume / cm ³	Surface area to volume ratio
C	_____	64	_____ : 1
D	_____	64	_____ : 1

[2 marks]



0 5 . 2 The student put the blocks into an acidic solution.

The acidic solution caused the blocks to gradually turn from pink to colourless.

She recorded the time taken for the blocks to turn completely colourless.
She repeated this three times.

Table 3 shows the student's results.

Table 3

Shape	Time for block to turn colourless / s				Mean time for block to turn colourless / s
	Block 1	Block 2	Block 3	Block 4	
C	3490	1200	3540	3530	
D	1680	1500	1590	1610	1595

After collecting the data, the student noticed that shape **C**, block 2 was damaged.

Calculate the mean for shape **C**.

[1 mark]

0 5 . 3 Suggest what the student should have done when she saw that shape **C**, block 2 was damaged.

[1 mark]

0 5 . 4 State **three** variables the student controlled in order to obtain valid results. **[2 marks]**

1 _____

2 _____

3 _____

Question 5 continues on the next page

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0 5 . 5

Describe how gas exchange occurs in single-celled organisms **and** explain why this method cannot be used by large, multicellular organisms.

[3 marks]

9

